

Customer Intelligence Pipeline

AAI5025 Final Project Report

Instructions

This report should summarize your findings from the Customer Intelligence Pipeline project, which involves cleaning raw data extracted from a database, creating new predictive features using NLP, and applying machine learning models (Linear Regression and K-Means) to understand user behavior and sentiment.

Grading Methodology Note

Important: This project is graded comparatively. There is no single "correct" answer that guarantees a perfect score. Grades for tasks such as model performance (R^2 Score, Accuracy) and code efficiency will be assigned relative to the performance of your peers. The students who achieve the most effective results and demonstrate the cleanest implementation will receive the highest scores.

1 Data Acquisition and Preparation

Cleaning Methodology

Explain the process used to prepare the raw data for analysis.

1. **Data Extraction (SQL):** We connected to the `user_data.db` database and executed a SELECT query to pull the `customer_reviews` table.
2. **Data Quality Issues:** The dataset contained two primary issues: inconsistent currency strings in the `Raw_Income` column (e.g., \$20,000) and missing values (NaNs) in the `Education` column.
3. **Cleaning Solution:** Explain the string manipulation used to clean the `Income` column and the imputation method used for `Education`.

Feature Engineering

Explain the two derived features created in the project.

- **Spend per Visit:** Explain why this feature was created and how it relates to customer loyalty.
- **Sentiment Score:** Explain that this score (Polarity) was generated using the `Review Text` and an NLP library, giving a numerical value to customer opinion.

2 Exploratory Data Analysis (EDA) and Visualization

This section visually presents key insights from the data using the visualizations created in the notebook.

Income Distribution

State your observation from the Histogram (Task C.1). Was the income distribution normal, or was it skewed? What does this mean for the "average" customer?

Comparison: Education vs. Spending

State your observation from the Bar Chart (Task C.3). Which education level showed the highest/lowest average `Spend_per_Visit`? Why might this be?

NLP Correlation Check

State your observation from the Scatter Plot (Task C.2) regarding the relationship between Income and Sentiment Score. Did higher income lead to more positive or negative reviews, or was the correlation weak?

Specific Observation: State the calculated correlation coefficient between `Review_Word_Count` and `Sentiment_Score` (Task B.2). What does this number imply about the length of a review and its overall emotion?

3 Predictive Modeling Results

Linear Regression (Predicting Spend)

The model was trained to predict `Total_Spend` based on `Clean_Income` and `Avg_Visits` (Task D.1). Use the following interpretation template to report the final R^2 Score and interpret your findings.

An R^2 score of [Your Score] means that [Your Score]% of the variation in customer spending is explained by their income and visit frequency. This suggests the model is [State whether the model is a strong or weak predictor based on the score].

K-Means Clustering (User Segmentation)

The K-Means model segmented the users into three distinct groups ($K = 3$). Describe the clusters visually observed in your scatter plot (Task D.2). What defines each segment? For example, Segment 1 might be 'High Income, Low Spend,' and Segment 3 might be 'High Spend, High Visits.'

- **Segment 0 (e.g., Budget Buyers):** Brief description of this group's characteristics.
- **Segment 1 (e.g., Loyal Spenders):** Brief description of this group's characteristics.

Classification: Predicting Gender (KNN)

The KNN model was trained to predict the Gender of the user based on Income and Total_Spend (Task D.3). Report the final Accuracy Score using the following template to conclude.

Conclusion: An Accuracy of [Your Score] indicates that the model is [State whether the model is good or bad] at predicting gender based on just these two features.